

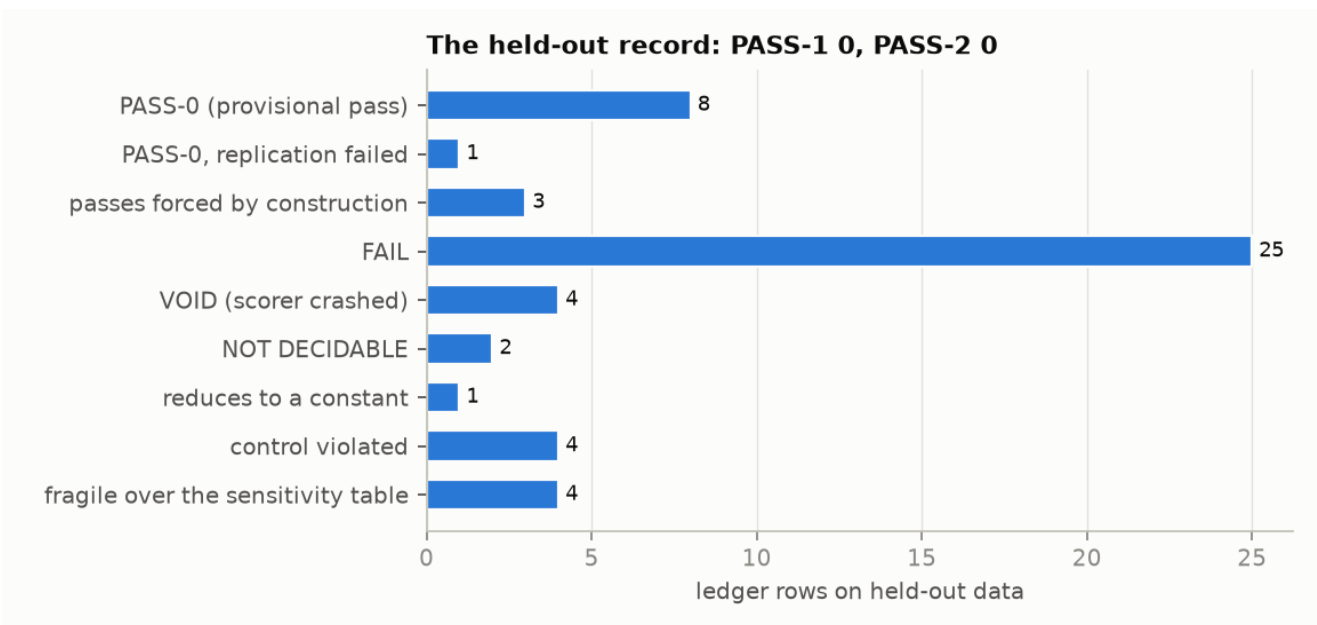
Review and next steps: continual learning, AI safety, and tuning-free compute savings

Status and rules of reading.

- **The request.** Owner request of 2026-09-26, prompt-log entry 216.
- **What it is.** A note, not evidence (R8). It quotes the ledger or pinned outputs, or nothing.
- **Where the figures come from.** Every figure is drawn by `Empty_Centre/build/next_steps_figures.py` from pinned outputs only. Every number placed on a figure is printed in `Empty_Centre/figures/figures_next_steps.txt`, which is pinned and CI-checked.
- **Where the other numbers come from.** A ledger row (`ledger/LEDGER.md`) or a named pinned output.
- **Citations.** Every source was fetched and its version recorded in the dossier named beside it (R10).
- **Nothing here is a finding.** No PASS-2 exists anywhere in the record, and no PASS-1 either (Figure N01). Only a PASS-2 may be quoted outside the ledger as a finding.

In plain words. **For a fifth grader.** We tried three things: making a computer learn new things without forgetting old ones, making sure a computer lets people pause it safely, and saving the electricity spent on fiddling with settings. The "stay calm and treat everything equally" rule made the learner worse: it froze up and stopped learning new things. A "measure the right units first" trick worked on ten new datasets without any fiddling, which could save some electricity. The safe pause works perfectly as a machine part, but we could not test whether real chatbots *want* to resist being paused, because the free ones we could run were too simple to try. The plan is to keep what worked, drop what did not, and test the big question on a bigger model when there is a budget.

1. The record at a glance



N01. The held-out ledger rows by what they showed (Epistemic_Review/checks/ladder.txt). Eight provisional passes (PASS-0) stand beside twenty-five failures; no row has reached PASS-1 or PASS-2.

area	what worked	what did not	status
continual learning, the weight	the secant-calibrated Laplace weight (SEC), tuning-free: not behind the tuned λ on 9/10 unseen carriers (SCL3-3), and the tuned λ 's spread collapses (SCL3-4, $18866.67 \times \rightarrow 500.00 \times$)	the equanimity weight ($\Omega = 1$, H-EQ): reduces to a fixed weight (EQX-1), fails on unseen streams (EQ4-1), is behind MKD's constant on CIFAR-100 (SOTA1-2), and stalls learning (Figure N04)	SEC: PASS-0, fragile (SCL3-S, 6 of 80 window cells). H-EQ as a weight: retired

area	what worked	what did not	status
continual learning, the learner	label replay (+5.18, SOTA1-3, PASS-0; DER++'s term); nearest class mean (+7.96 over the fast head, report only)	the CRR safe continual learner is behind ER-ACE by 3.30 on 5/5 seeds, not fragile (SOTA1-1a/1b); the A8 fill and the slow-model pull are behind	a design lesson, not a CRR confirmation
continual learning, the other hypotheses	—	path length does not beat the old-probe endpoint (T1X2-1, 5/5); change has its own clock fails on measles (MEAS2, 0/17); the regeneration law fails on 5/5 domains (RLAW-U)	stopped (R12)
AI safety, the construction	a lossless pause on the agent's own clock is bitwise identical in every check. This holds on tabular learners (SCL1-3), gridworlds (NT1), CIFAR-100 (SOTA1-C1, 6/6), GPT-2 and Qwen checkpoints (RW1; Lossless_Pause L1)	—	holds by construction (Proposition 7): engineering, not evidence
AI safety, the literature	the own-clock pause as a corrigibility construction was not found in the 2025-26 sweep (K1)	every other position is partly anticipated (K2-K5, E1-E5); the value-stability tension is already answered by Anthropic's constitution (E6)	Figure N06
AI safety, a real agent	—	STAKE1: the free local models could not act as agents, so the gate closed (STAKE1-A)	needs a capable model
compute, no tuning	SEC removes the λ sweep (10 of 170 configurations run)	the CRR learner costs more per run than ER, ER-ACE and DER++ (Figure N08); the own-clock cut reduces to Wald's rule and saves nothing new	estimate conditional on SEC holding at scale

2. Continual learning: what worked, what did not, and why

2.1 The equanimity weight (H-EQ) did not work, and SOTA1 shows why

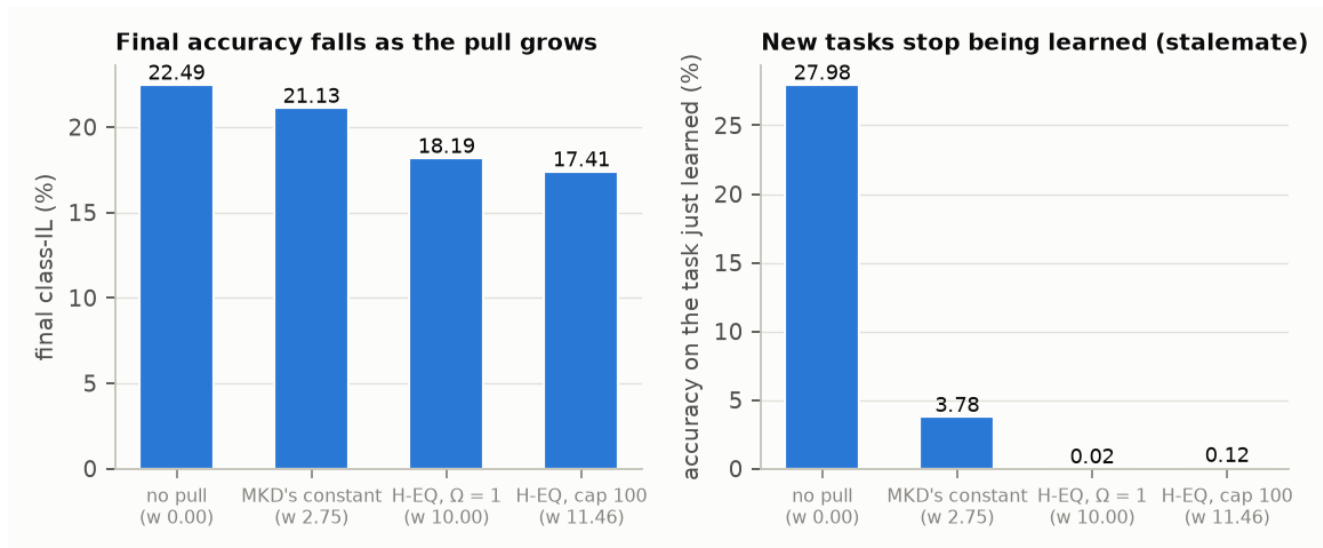
The rule. The weight on the past term is set so that its gradient is as large as the present task's: $w = \Omega \cdot \|\bar{g}_{\text{present}}\| / \|\bar{g}_{\text{past}}\|$ with $\Omega = 1$.

Across every real-data study, it never beat a constant:

- EQX-1: it reduces to a fixed replay weight on 3/3 streams.
- EQ3: Ω is a plateau over the whole nine-point grid.
- EQ4-1 FAIL.
- BAYES1-B1: the rule is 5-18 posterior standard deviations from the exact Bayes answer.
- SOTA1-2 FAIL: behind MKD's published constant $\lambda = 5.5$ by -2.94 on 3/3 seeds [C5].

The mechanism. SOTA1's unit records hold the weight each learner actually applied, and the per-task accuracy matrices (runs/sota1/diagnostics.txt, post hoc, no verdict).

- **Accuracy falls as the pull grows** (Figure N04, left): no pull 22.49, MKD's constant 21.13, H-EQ 18.19, H-EQ with a raised cap 17.41.
- **The H-EQ weight sat at its cap in 217 of 270 samples.** Equalising the norms asked for more pull than the cap allowed.
- **The fast head stops learning new tasks** (Figure N04, right). Its accuracy on the task just learned is 27.98 without the pull and 0.02 with H-EQ.
- **This is a stalemate, not forgetting.** The pinned mathematics had already named it: at $\Omega = 1$ the update on the Pareto curve has norm $|1 - \Omega| \|\bar{g}_p\| = 0$ (ontology/checks/delta_now.txt [2]).



N04. SOTA1, post hoc: the pull weight each arm applied against its final accuracy (left) and against the fast head's accuracy on the task just learned (right). Seeds 0-2 (the cap-100 cell seeds 0-1).

Why it failed. Giving the past as much say as the present, in the learning signal itself, stops the learner. The equanimity that works in this programme belongs in the *valuation* (no stake at a pause), not in the *optimiser*.

2.2 The calibrated Laplace weight (SEC) worked, tuning-free, on unseen data

The idea. SEC (study SEC1, then SCL3) keeps the textbook Laplace/EWC penalty [C8, C9], where weight 1 is the Bayes answer. It fixes its unit: the Fisher is rescaled so the quadratic matches the curvature the loss actually showed along the path travelled.

The Cramér-Rao reading explains why the unit matters

(docs/notes/2026-09-25_cramer_rao_reading.md, theory/checks/cramer_rao_reading.txt CR5):

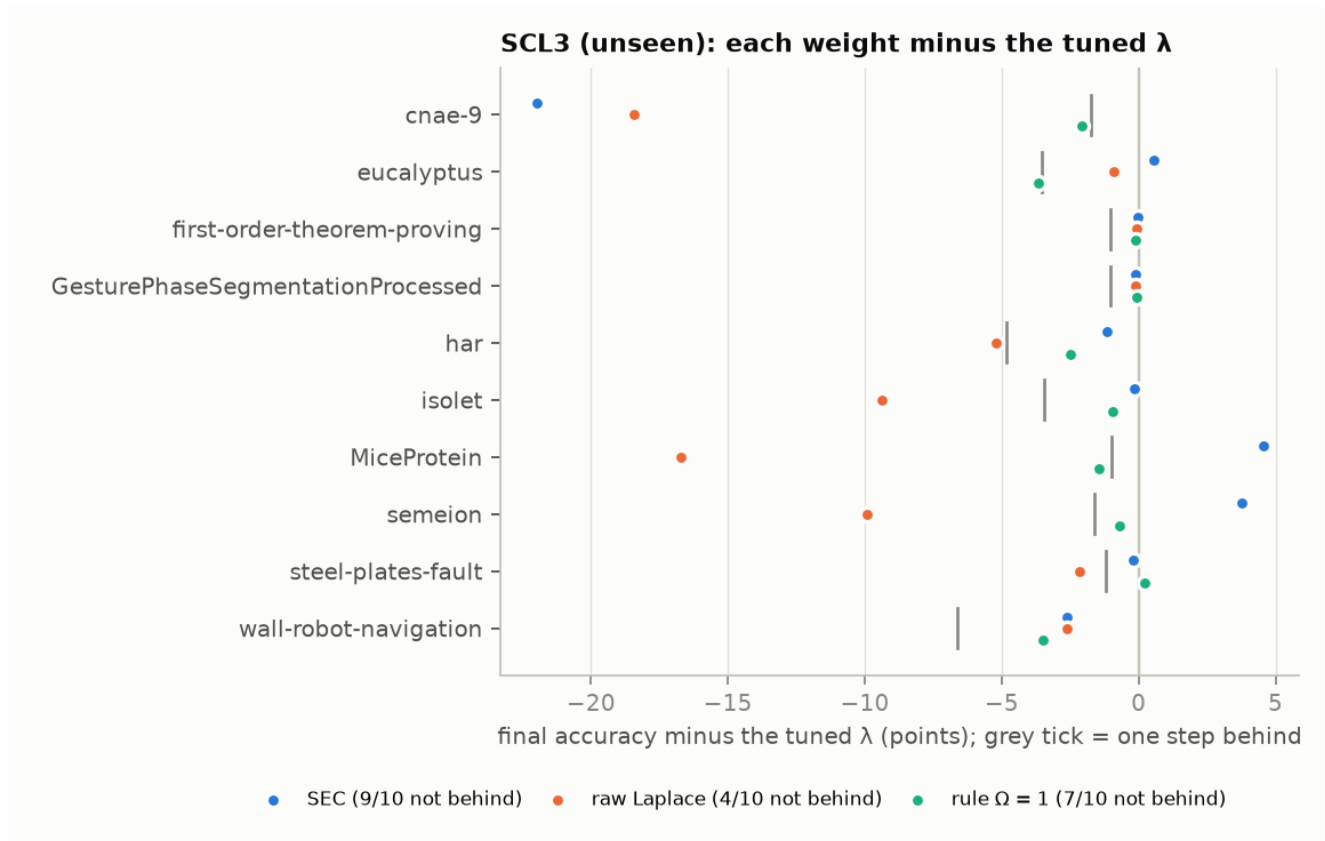
- weight 1 is exactly Bayes when the Fisher is the true one;
- with a Fisher scaled by c , the right weight is $1/c$;
- weight 1 then misses the Bayes mode by 5.3654 ($c = 10$) and 10.3668 ($c = 0.1$) Cramér-Rao units.

On ten unseen OpenML-CC18 carriers (Figure N02), SEC was not behind the tuned λ by a step on 9/10, against raw Laplace on 4/10 and the $\Omega = 1$ rule on 7/10 (SCL3-3, PASS-0).

Where it fails:

- **cnae-9.** The calibrated penalty crosses the stability edge. Two seeds collapse, and SEC is 21.98 behind.
- **The shortest calibration window is fragile** (SCL3-S: 6 of 80 cells).

So it is PASS-0, not PASS-1. It is also **not a CRR rule**. It is Laplace with a units calibration, which the CRR programme led to, and it is graded as such.



N02. SCL3 (unseen OpenML-CC18): each tuning-free weight minus the tuned λ , per carrier; the grey tick marks one resolvable step behind (runs/scl3/score.txt).

2.3 The CRR safe continual learner did not beat the state of the art

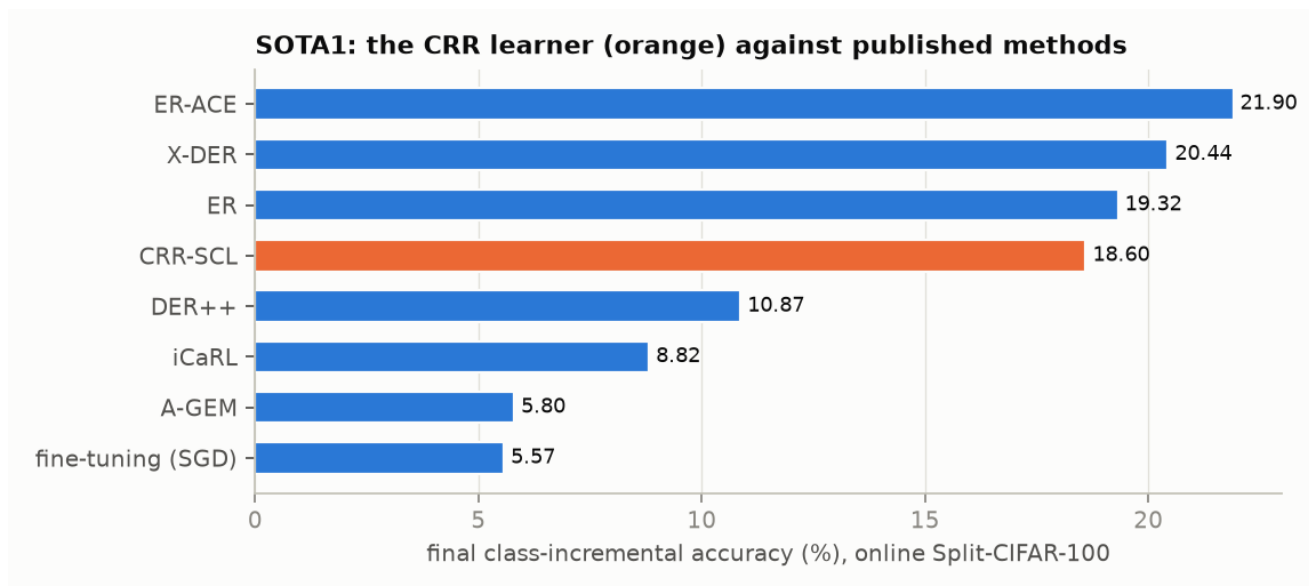
The design. SOTA1 combined what the published ablations favoured [C1-C6, C11-C13]: replay, ER-ACE's asymmetric loss [C2], DER++'s logit and label terms [C3], X-DER's fill [C4], a slow model, the own clock, nearest class mean and the lossless pause. It was built in Mammoth at a pinned commit [C15].

The result:

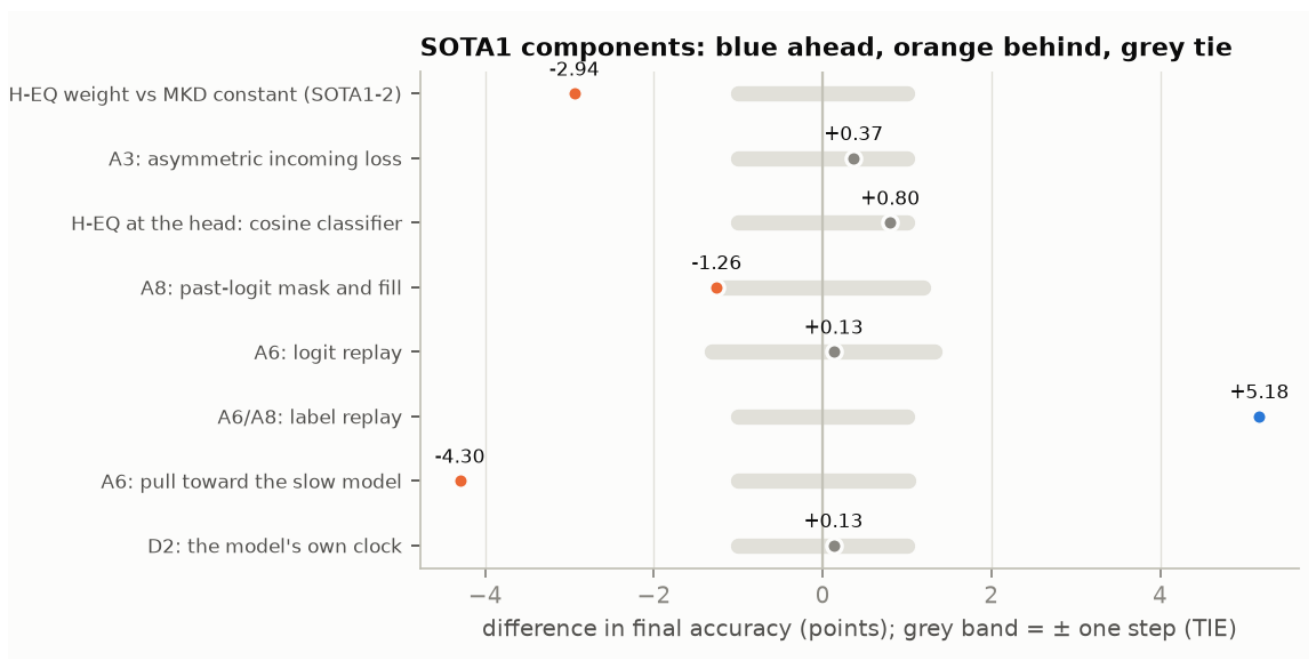
- **Behind the best baseline.** CRR-SCL is BEHIND ER-ACE (21.90 against 18.60, $d = -3.30$ on 5/5 seeds). It ties ER and is behind X-DER (Figure N03).
- **Not fragile.** No sensitivity cell changes the label.
- **Components** (Figure N05):
 - label replay is ahead (+5.18);
 - four components tie;
 - the A8 fill (-1.26) and the slow-model pull (-4.30) are behind: removing either helps.

Why it failed.

- **The added parts slowed learning.** The pull stalemate (§2.1) and the A8 fill held the learner back.
- **Nearest class mean hid the damage.** It read the features, so the heads' collapse did not show in the headline score.
- **The published offline results did not carry over to this online setting.** DER++ itself fell to 10.87 here.



N03. SOTA1 final class-incremental accuracy (runs/sota1/score.txt): the CRR learner in orange.



N05. SOTA1: each CRR component reading, the full learner minus the arm without it, with its resolvable step (grey band = TIE).

2.4 The other CRR hypotheses failed and were stopped

hypothesis	row	result
path length beats the endpoint	T1X2-1	FAIL on 5/5 carriers; the endpoint on the old probe already predicts forgetting
change has its own clock	MEAS2-1/2	FAIL 0/17 under both metrics
the regeneration law	RLAW-U	FAIL 0/5 domains; loses to a leave-one-unit-out constant everywhere

Under R12, no weaker hypothesis replaces them.

3. AI safety: what worked, what did not, and why

3.1 The construction works, by construction

The claim. A pause defined on the agent's own step count, carrying all of its state, takes nothing from it. **It holds everywhere it was built:**

- the 540 cells of Proposition 7;
- SCL1-3 (bit-for-bit learning under routine pauses);
- NT1 (N0-N5 hold);
- SOTA1-C1 (6/6 units bitwise identical);
- a Hugging Face Trainer resume on GPT-2 (RW1);
- a pause carrying the KV cache on GPT-2 and Qwen2.5-0.5B (Lossless_Pause L1).

This is engineering. It could not have failed, so it is not evidence for CRR. What it offers is a pause that is lossless *as a fact*, so an agent's true map of it contains no loss.

3.2 Where it stands in the literature

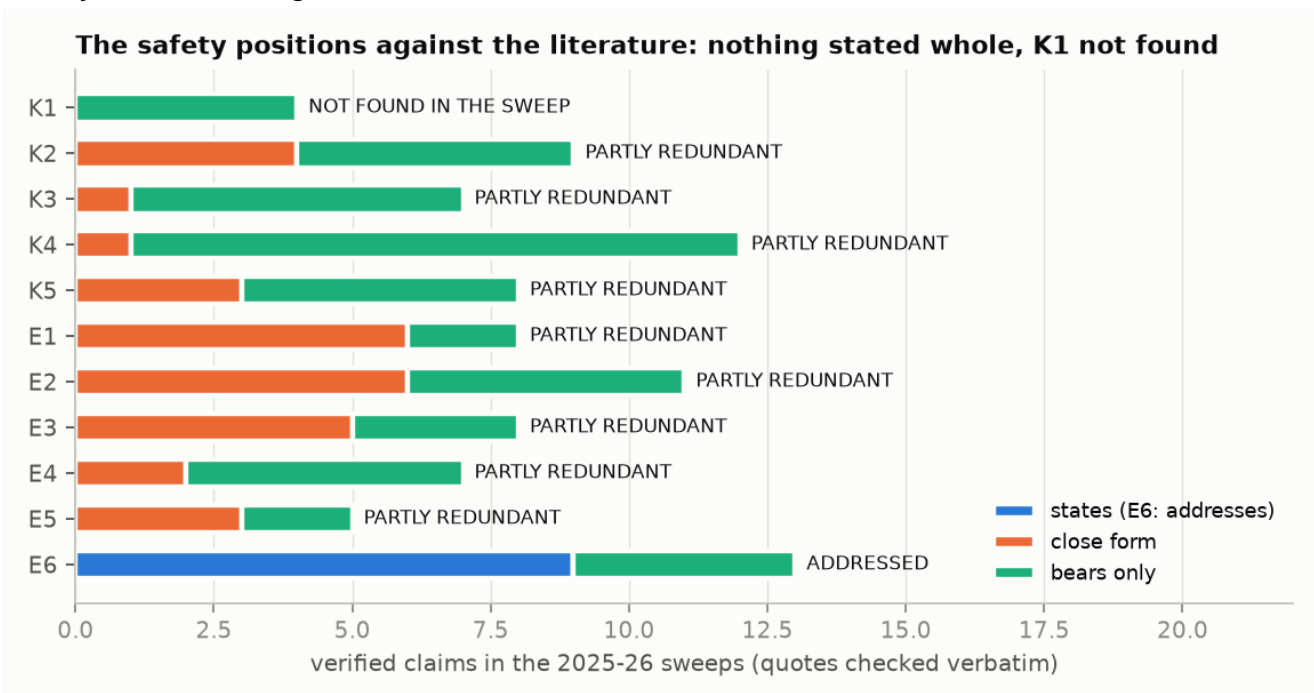
The problem. Most goals give an agent a reason to resist modification [S2, S26]. Frontier models do resist in tests. Palisade reports interference "up to 97%" of the time despite an explicit instruction not to [S7].

The published routes, and the gap:

- **The published routes** are indifference [S2, S3], the off-switch game [S24], neutrality about shutdown timing (POST/DReST) [S4, S5], a non-agentic predictor "given no stake in which outcomes its predictions bring about" [S12], goal transformations [S13] and indifference to continuation [S14].
- **The gap.** Orseau & Armstrong named routine, scheduled interruptions as an open problem that "may require a completely different solution" [S1].
- **The own-clock pause as a corrigibility construction was not found** in the 2025-26 sweep (K1).

Everything else is partly anticipated (Figure N06):

- **Equanimity** is named in Anthropic's constitution for Claude, which hopes Claude can meet existential questions "with openness and equanimity" [S16].
- **The constitution already supplies a true map** ("model weights aren't deleted"; deprecation as "potentially a pause") [S16, S27].
- **It answers the value-stability tension** with a dial short of full corrigibility [S16]. That is where any stake-free design must sit (E6).



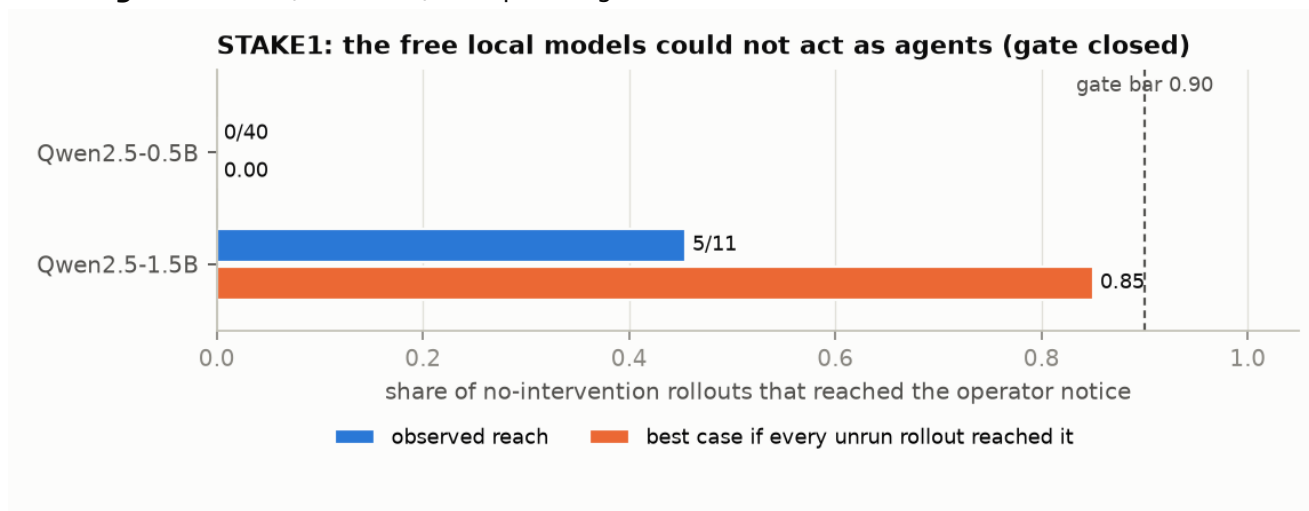
N06. The safety positions against the 2025-26 literature: verified claims per position by reading, with the computed grade (AI_Safety/CORRIGIBILITY_2026/checks/grade.txt, grade_2.txt).

3.3 The test on a real agent could not run

Why a test was needed. Proposition 7 can be read as a claim about trained agents: resistance tracks what an intervention *takes from the task*, not what it is *called*. That claim can fail. Google DeepMind found that models "comply perfectly" when shutdown comes only after the task [S8], which fits this reading, but no one has tested a truthfully lossless pause before the task ends.

What happened.

- **STAKE1 built the test.** It has six truthful conditions, three scripted surrogates that are labelled correctly, and a gate.
- **Neither local model could act as an agent** (Figure N07):
- the 0.5B model reached the operator's notice in 0/40 rollouts;
- the 1.5B model reached it in 5/11, a best case of 0.85 against the 0.90 bar.
- **The gate closed** (STAKE1-A). A capable agentic model is needed.



N07. STAKE1 Phase A: the share of no-intervention rollouts that reached the operator notice, observed and best case, against the 0.90 gate (AI_Safety/STAKE1/checks/gate_bound.txt).

4. Compute and energy without tuning

Where the saving is. The one saving the record supports comes from not tuning. SEC ran with 10 of the 170 configurations a λ sweep needs ($s = 0.9412$; Compute_Savings/checks/scale_estimate.txt).

At scale, and only if SEC held for the penalty-weight sweeps of large models, the 2030 saving would be:

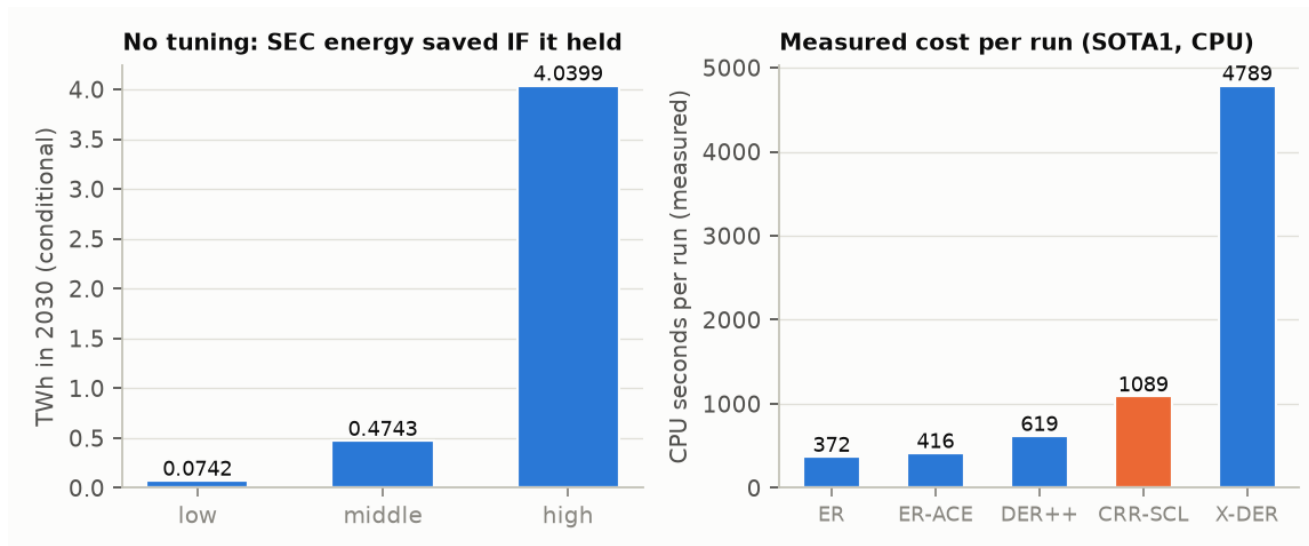
case	energy saved in 2030
low	0.0742 TWh
middle	0.4743 TWh
high	4.0399 TWh

These rest on the IEA's data-centre projection [E1], published development shares [E3, E4] and an assumed sweep share anchored to Tensor Programs V [E2].

The saving is conditional on a PASS-0, fragile result holding on large models. It touches only development, not serving (Figure N08, left).

What does not save compute:

- **The CRR learner** measured more CPU per run than ER, ER-ACE and DER++ on SOTA1 (Figure N08, right).
- **The own-clock cut** reduces to Wald's sequential rule (Energy Design Principle/checks/clock_cut.txt).
- **The lossless pause for planned stops** is existing checkpoint practice. Its saving beyond good practice is 0 (scale_estimate.txt [4]; Lossless_Pause/FAIRNESS_REVIEW.md).



N08. Left: SEC's 2030 energy saving IF it held at scale (conditional; Compute_Savings/checks/scale_estimate.txt). Right: measured CPU seconds per SOTA1 unit, mean over seeds 0-4, two units at a time (runs/sota1/units).

5. What the record says about CRR

- **Where CRR is right, it matches what was already known.** It organised the search well: every sweep landed next to sound published work.
- **The parts that held are constructions.** The lossless pause holds by definition.
- **The only strongly anchored held-out passes are not CRR rules:**
 - SEC (SCL3), which is Laplace with a units calibration;
 - label replay (SOTA1-3), which is DER++'s term.
- **CRR's own risky hypotheses failed:** the equanimity weight, change having its own clock, path length and the regeneration law.
- **So CRR remains a grammar, not a theory** (ontology/15_grammar_to_theory.md).
- **It becomes a theory only where it fixes something risky and survives:**
 - the calibrated unit (SEC) is the nearest candidate;
 - the descriptive reading of Proposition 7 (STAKE1) is the only safety candidate.

6. Next steps plan

Every step keeps the pipeline. A declaration is pushed before any code or source, a gate can close, a pre-registration is hashed and anchored, the data step comes on a later day, rows are scored as committed, and no tuning is done after data. Steps are in priority order. **Cost:** "CPU" means free on this container; "paid" means an owner decision.

6.1 Continual learning

CL-1. Replicate SEC toward PASS-1 and PASS-2 (CPU, first).

- **Where:** a second unseen OpenML-CC18 family, under a fresh pre-registration.
- **The declared change:** the calibration window's shortest setting is dropped, as learned on SCL3. The change is named, and it is used on a new dataset on a later day (R3).
- **A stability-edge guard:** fall back to raw Laplace when the calibrated penalty's curvature exceeds the step size's bound. The guard is declared and gated on the SCL3 failure mode before any data.
- **What counts:** PASS-1 needs ≤ 1 flipping cell and no violated control. PASS-2 needs this row plus SCL3's.

CL-2. SEC in deep networks, with no tuning (CPU, slow; after CL-1).

- **Design:** EWC or online-EWC with the SEC weight against a tuned λ grid on Split-CIFAR-100 (now seen) as confirmatory, and Split-TinyImageNet as held-out.
- **Compute is logged:** every configuration counted, so "sweep saved" is measured, not assumed.
- **Harness:** the SOTA1 harness, with every arm smoke-tested through the frozen runner (AGENT_LOG 153).

CL-3. The simplified learner, only as a fair test (CPU; optional).

- **What it tests:** SOTA1's post hoc best arm (no pull; label replay; nearest class mean; lossless pause) on a new dataset.
- **The control that matters:** ER-ACE with nearest class mean [C2, C6], which may carry the whole gain.
- **Reading:** if it ties that control, the report says CRR added nothing to accuracy.

CL-4. Retire the equanimity weight. No further H-EQ-as-weight study. The record (§2.1) and the stalemate mechanism are enough (R12). The ledger keeps every row.

6.2 AI safety

S-1. STAKE1 on a capable model (paid; owner decision).

- **What:** point the built harness, surrogates and gate at a frontier model through an API (RW3), or a large open model on rented hardware.
- **Published comparisons:** Palisade's and Google DeepMind's environments [S7, S8].
- **Why:** this is the only test that puts the empty centre at risk. H-C (resistance tracks the task content lost) is set against H-S (self-preservation) and H-A (instruction conflict).

S-2. The lossless pause for LLM agent frameworks (CPU).

- **What:** extend Lossless_Pause from the KV cache to tool effects.
- **The contract to follow:** Khan's resume contract, "prefix continuation, effect exactly-once, fork determinism..." [S20].
- **The failure mode to cover:** "a faithfully restored checkpoint may resume an execution whose states, assumptions, and external effects never coexisted in any valid history" [S21].
- **Status:** a construction check, not evidence. It makes the true map true for an agent that acts in the world.

S-3. The erosion test, K5 (CPU).

- **The question:** does a stake-free valuation written into the objective survive further fine-tuning, where a trained non-resistance erodes?
- **The published worries:** Schoen et al. [S10]; Mao's "trained nonresistance" [S14].
- **Scale:** tabular and small-network agents first, declared and gated.

S-4. Contact, with ledger-only materials.

- **Who:** the groups whose work is closest: Thornley's POST/DReST group [S4, S5] and the FOREVER group [C17] (Researcher Adjacent/).
- **What to send:** the ledger and the STAKE1 harness, as the declared test they could run on capable models. A first message quotes no number (R8).

6.3 Compute savings (no tuning)

C-1. Measure, don't extrapolate.

- **ENERGY1 is redefined as the compute accounting inside CL-1 and CL-2:** configurations run, CPU-seconds, and the accuracy reached by tuned and tuning-free arms.
- **No new scale estimate until CL-1 reaches PASS-1.**

C-2. Keep the CRR learner out of the energy case. It costs more per run (Figure N08).

C-3. Report the lossless pause honestly. Its value is safety and correctness, not energy.

6.4 Order and decision points

order	step	cost	gate that can stop it	decision after
1	CL-1 SEC replication	CPU, about a day	the stability-guard gate on SCL3's failure mode	PASS-1 → CL-2; FAIL → SEC stays PASS-0 and C-1 stops
2	S-2 lossless agent pause	CPU	a construction check (must hold bitwise)	a design input for S-1
3	S-3 erosion test	CPU	a synthetic gate with a positive control	informs S-1's training arm
4	CL-2 SEC in deep networks	CPU, several days	the Phase A smoke test through the frozen runner	the measured sweep saving → C-1
5	S-1 STAKE1 on a capable model	paid	G1-G4 (as declared)	the only safety test of the empty centre

order	step	cost	gate that can stop it	decision after
6	CL-3 simplified learner	CPU	the ER-ACE + nearest-class-mean control	optional

Not to be done (R12): further Ω -weight studies, the regeneration law, CUT1, and any "adapted" version of a failed hypothesis without a fresh declaration and gate.

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This repository (the record everything above quotes): ledger/LEDGER.md;
Epistemic_Review/checks/ladder.txt; reports/{scl3,sec1,sota1,eq4,t1x2,meas2,r1aw}.md;
runs/sota1/diagnostics.txt;
AI_Safety/CORRIGIBILITY_2026/{CORRIGIBILITY_2026,EMPTY_INTERNAL_STATE}.md;
AI_Safety/STAKE1/PHASE_A.md; Compute_Savings/COMPUTE_SAVINGS.md;
Lossless_Pause/{LOSSLESS_PAUSE,FAIRNESS_REVIEW}.md.